

AI-Assisted Coding LAB TEST_4

Roll Number:2403A51457

SET:1

Q2.

Use AI to generate a Python script that:

- Randomly creates a numeric dataset representing daily website traffic
- Identifies and removes outliers using the IQR (Interquartile Range) method
- Exports both the original and cleaned datasets to two separate CSV files

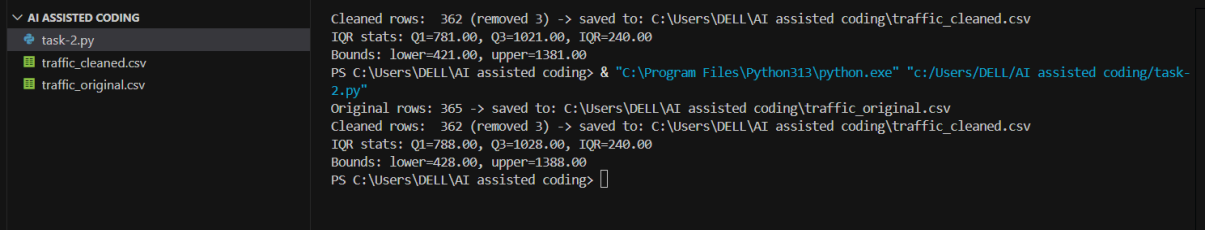
PROMPT:

generate a Python script that:

- Randomly creates a numeric dataset representing daily website traffic
- Identifies and removes outliers using the IQR (Interquartile Range) method
- Exports both the original and cleaned datasets to two separate CSV files.

```
1 # ...existing code...
2 import os
3 import argparse
4 from datetime import datetime, timedelta
5
6 import numpy as np
7 import pandas as pd
8
9 def generate_traffic(days=365, seed=None):
10     rng = np.random.default_rng(seed)
11     end = pd.Timestamp.today().normalize()
12     dates = pd.date_range(end=end, periods=days, freq='D')
13     # base seasonality (weekly) + mild upward trend
14     day_of_week = dates.dayofweek.values
15     base = 800 + 150 * np.sin(2 * np.pi * day_of_week / 7) + np.linspace(0, 200, days)
16     noise = rng.normal(loc=0, scale=120, size=days)
17     visits = np.round(base + noise).astype(int)
18     # inject random spikes and dips (outliers)
19     n_spikes = max(3, days // 120)
20     spike_idx = rng.choice(days, size=n_spikes, replace=False)
21     for i in spike_idx:
22         # either a big spike or a deep dip
23         visits[i] += int(rng.choice((-1500, -400, 2000)))
24     visits = np.where(visits < 0, 0, visits) # no negative visits
25     df = pd.DataFrame({'date': dates, 'visits': visits})
26
27 # Cleaned rows: 362 (removed 3) -> saved to: C:\Users\DELL\AI assisted coding\traffic_cleaned.csv
28 IQR stats: Q1=781.00, Q3=1021.00, IQR=240.00
29 Bounds: lower=421.00, upper=1381.00
30 PS C:\Users\DELL\AI assisted coding> & "C:\Program Files\Python313\python.exe" "C:\Users\DELL\AI assisted coding/task-2.py"
31 Original rows: 365 -> saved to: C:\Users\DELL\AI assisted coding\traffic_original.csv
32 Cleaned rows: 362 (removed 3) -> saved to: C:\Users\DELL\AI assisted coding\traffic_cleaned.csv
33 IQR stats: Q1=788.00, Q3=1028.00, IQR=240.00
34 Bounds: lower=428.00, upper=1388.00
35 PS C:\Users\DELL\AI assisted coding>
```

OUTPUT:



```
AI ASSISTED CODING
task-2.py
traffic_cleaned.csv
traffic_original.csv

Cleaned rows: 362 (removed 3) -> saved to: C:\Users\DELL\AI assisted coding\traffic_cleaned.csv
IQR stats: Q1=781.00, Q3=1021.00, IQR=240.00
Bounds: lower=421.00, upper=1381.00
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IQR stats: Q1=788.00, Q3=1028.00, IQR=240.00
Bounds: lower=428.00, upper=1388.00
PS C:\Users\DELL\AI assisted coding>
```

Q1.

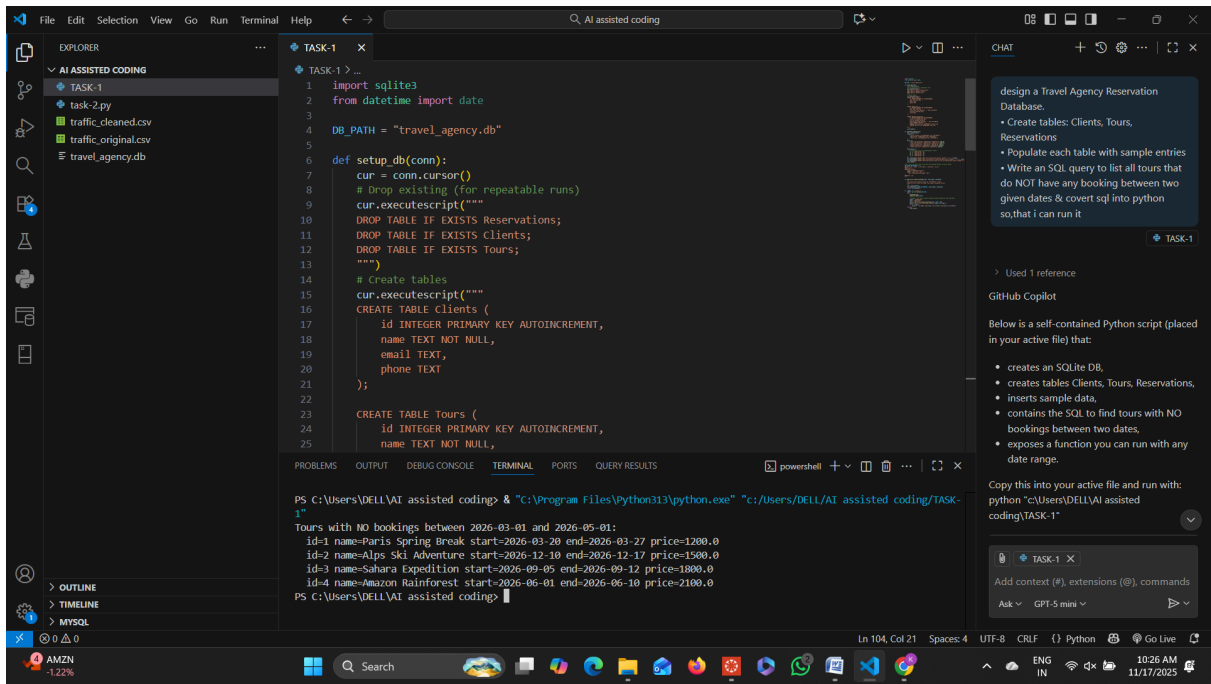
Using AI assistance, design a Travel Agency Reservation Database.

- Create tables: Clients, Tours, Reservations
- Populate each table with sample entries
- Write an SQL query to list all tours that do NOT have any booking between two given dates

PROMPT:

design a Travel Agency Reservation Database.

- Create tables: Clients, Tours, Reservations
- Populate each table with sample entries
- Write an SQL query to list all tours that do NOT have any booking between two given dates & covert sql into python



OUTPUT:

